Calculus I

Solution to Assignment 7

1. To find the interval where the function is covex, we need to find the interval on which the function's second derivative is positive.

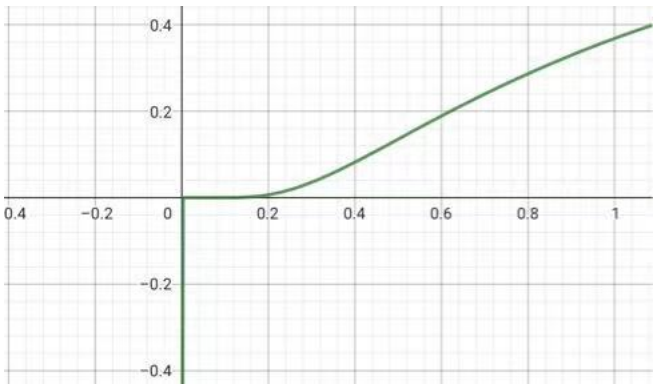
   (a) The domain of $f(x)$ is $\mathbb{R}$. $f'(x) = 20x^3 - 9x^2 + 2x$, $f''(x) = 60x^2 - 18x + 2$. Solve $f''(x) > 0$. Since $a = 60 > 0$ and $\Delta = 18^2 - 4 \times 60 \times 2 = -156 < 0$, $f''(x) > 0$ for all $x \in \mathbb{R}$. The function is convex on $\mathbb{R}$.

   (b) The domain of $f(x)$ is $(-\infty, 1) \cup (1, \infty)$. $f'(x) = -2(x-1)^{-2}$, $f''(x) = 4(x-1)^{-3}$. $f''(x) > 0$ for $x > 1$, so the functionis convex on $(1, \infty)$.

   (c) The domain of $f(x)$ is $[0, \infty)$. $f'(x) = \frac{1}{2}x^{-1/2}$, $f''(x) = -\frac{1}{4}x^{-3/2}$. Since $f''(x)$ is always negative for $x > 0$, the interval where the function is convex does not exist.

   (d) The domain of the $f(x)$ is $(0, \infty)$. $f'(x) = -\frac{1}{2}x^{-3/2}$, $f''(x) = \frac{3}{4}x^{-5/2}$. Since $f''(x)$ is always positive for $x > 0$, the function is convex on $(0, \infty)$.

2. The domain of $e^{-1/x}$ is $(-\infty, 0) \cup (0, \infty)$. $(e^{-1/x})' = e^{-1/x}x^{-2}$, $(e^{-1/x})'' = e^{-1/x}x^{-4}(1 - 2x)$. $(e^{-1/x})'' > 0$ if $x < 0$ or $0 < x < \frac{1}{2}$. Then $e^{-1/x}$ is convex on $(-\infty, 0)$ and $(0, \frac{1}{2})$. So $b = \frac{1}{2}$. The graph on $(0, \infty)$ is as below.



3.

(a) This limit has the form $\frac{0}{0}$. So $\lim_{x\to\pi/4}\frac{\sin x-\cos x}{\tan x-1}\overset{H}{=}\lim_{x\to\pi/4}\frac{\cos x+\sin x}{\sec^2 x}=\lim_{x\to 4}\frac{\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}}{(\sqrt{2})^2}=\frac{\sqrt{2}}{2}$

(b) This limit has the form $\frac{0}{0}$.

$$\lim_{x\to 0}\frac{\sqrt{1+2x}-\sqrt{1-4x}}{x}\overset{\text{H}}{=}\lim_{x\to 0}\frac{\frac{1}{2}(1+2x)^{-1/2}\cdot 2-\frac{1}{2}(1-4x)^{-1/2}(-4)}{1}$$
$$=\lim_{x\to 0}\left(\frac{1}{\sqrt{1+2x}}+\frac{2}{\sqrt{1-4x}}\right)=\frac{1}{\sqrt{1}}+\frac{2}{\sqrt{1}}=3$$

(c) This limit has the form $\frac{0}{0}$. $\lim_{x\to 1}\frac{x\sin(x-1)}{2x^2-x-1}\overset{\text{H}}{=}\lim_{x\to 1}\frac{x\cos(x-1)+\sin(x-1)}{4x-1}=\frac{\cos 0}{4-1}=\frac{1}{3}$

(d) Note that $y=x^{\sqrt{x}}\Rightarrow\ln y=\sqrt{x}\ln x$. So

$$\lim_{x\to 0^+}\ln y=\lim_{x\to 0^+}\sqrt{x}\ln x=\lim_{x\to 0^+}\frac{\ln x}{x^{-1/2}}\overset{\text{H}}{=}\lim_{x\to 0^+}\frac{1/x}{-\frac{1}{2}x^{-3/2}}=-2\lim_{x\to 0^+}\sqrt{x}=0\Rightarrow$$
$$\lim_{x\to 0^+}x^{\sqrt{x}}=\lim_{x\to 0^+}e^{\ln y}=e^0=1$$

(e) Since $y=(1-\cos x)^{\sin x}\Rightarrow\ln y=\sin x\ln(1-\cos x)$, so

$$\lim_{x\to 0^+}\ln y=\lim_{x\to 0^+}\sin x\ln(1-\cos x)=\lim_{x\to 0^+}\frac{\ln(1-\cos x)}{\csc x}\overset{\text{H}}{=}\lim_{x\to 0^+}\frac{\frac{1}{1-\cos x}\cdot\sin x}{-\csc x\cot x}$$
$$=-\lim_{x\to 0^+}\frac{\sin x}{(1-\cos x)\csc x\cot x}=-\lim_{x\to 0^+}\frac{\sin x}{\csc x\cot x-\cot^2 x}\cdot\left(\frac{\sin^2 x}{\sin^2 x}\right)$$
$$=-\lim_{x\to 0^+}\frac{\sin^3 x}{\cos x-\cos^2 x}=-\lim_{x\to 0^+}\frac{\sin^3 x}{(1-\cos x)\cos x}$$
$$\overset{\text{H}}{=}-\lim_{x\to 0^+}\frac{3\sin^2 x\cos x}{(1-\cos x)(-\sin x)+\cos x(\sin x)}=-\lim_{x\to 0^+}\frac{3\sin x\cos x}{(\cos x-1)+\cos x}=-\frac{0}{0+1}=0.$$

And $\lim_{x\to 0^+}(1-\cos x)^{\sin x}=\lim_{x\to 0^+}e^{\ln y}=e^0=1$.

(f) Note that $y=\left(\frac{2x-3}{2x+5}\right)^{2x+1}\Rightarrow\ln y=(2x+1)\ln\left(\frac{2x-3}{2x+5}\right)\Rightarrow$

$$\lim_{x\to\infty}\ln y=\lim_{x\to\infty}\frac{\ln(2x-3)-\ln(2x+5)}{1/(2x+1)}\overset{\text{H}}{=}\lim_{x\to\infty}\frac{2/(2x-3)-2/(2x+5)}{-2/(2x+1)^2}$$
$$=\lim_{x\to\infty}\frac{-8(2x+1)^2}{(2x-3)(2x+5)}=\lim_{x\to\infty}\frac{-8(2+1/x)^2}{(2-3/x)(2+5/x)}=-8,$$

which implies $\lim_{x\to\infty}\left(\frac{2x-3}{2x+5}\right)^{2x+1}=\lim_{x\to\infty}e^{\ln y}=e^{-8}$.

4. The limit, $L = \lim_{x\to\infty}\left[x - x^2\ln\left(\frac{1+x}{x}\right)\right] = \lim_{x\to\infty}\left[x - x^2\ln\left(\frac{1}{x}+1\right)\right]$. Let $t = 1/x$, so as $x \to \infty, t \to 0^+$.

$$\begin{aligned} L &= \lim_{t\to 0^+}\left[\frac{1}{t} - \frac{1}{t^2}\ln(t+1)\right] = \lim_{t\to 0^+}\frac{t - \ln(t+1)}{t^2} \\ &\overset{\text{H}}{=} \lim_{t\to 0^+}\frac{1 - \frac{1}{t+1}}{2t} = \lim_{t\to 0^+}\frac{t/(t+1)}{2t} = \lim_{t\to 0^+}\frac{1}{2(t+1)} = \frac{1}{2} \end{aligned}$$

Note: Starting the solution by factoring out x or x^2 leads to a more complicated solution.

5. $L = \lim_{x\to 0}\left(\frac{\sin 2x}{x^3} + a + \frac{b}{x^2}\right) = \lim_{x\to 0}\frac{\sin 2x + ax^3 + bx}{x^3} \overset{H}{=} \lim_{x\to 0}\frac{2\cos 2x + 3ax^2 + b}{3x^2}$. As $x \to 0, 3x^2 \to 0$, and $\left(2\cos 2x + 3ax^2 + b\right) \to b + 2$, so the last limit exists only if $b + 2 = 0$, that is, $b = -2$. Thus, $\lim_{x\to 0}\frac{2\cos 2x + 3ax^2 - 2}{3x^2} \overset{\text{H}}{=} \lim_{x\to 0}\frac{-4\sin 2x + 6ax}{6x} \overset{\text{H}}{=} \lim_{x\to 0}\frac{-8\cos 2x + 6a}{6} = \frac{6a-8}{6}$, which is equal to 0 if and only if $a = \frac{4}{3}$. Hence, $L = 0$ if and only if $b = -2$ and $a = \frac{4}{3}$.

6.

(a) $\frac{d}{dx}[x\ln x - x + C] = x\cdot\frac{1}{x} + (\ln x)\cdot 1 - 1 + 0 = 1 + \ln x - 1 = \ln x$

(b) We have

$$\begin{aligned} &\frac{d}{dx}\left[\frac{2}{15b^2}(3bx - 2a)(a+bx)^{3/2} + C\right] \\ &= \frac{2}{15b^2}\left[(3bx - 2a)\frac{3}{2}(a+bx)^{1/2}(b) + (a+bx)^{3/2}(3b) + 0\right] \\ &= \frac{2}{15b^2}(3b)(a+bx)^{1/2}\left[(3bx - 2a)\frac{1}{2} + (a+bx)\right] \\ &= \frac{2}{5b}(a+bx)^{1/2}\left(\frac{5}{2}bx\right) = x\sqrt{a+bx} \end{aligned}$$

7.

(a) $$\begin{aligned} \int\sqrt{t}\left(t^2+3t+2\right)dt &= \int t^{1/2}\left(t^2+3t+2\right)dt = \int\left(t^{5/2} + 3t^{3/2} + 2t^{1/2}\right)dt \\ &= \frac{2}{7}t^{7/2} + 3\cdot\frac{2}{5}t^{5/2} + 2\cdot\frac{2}{3}t^{3/2} + C = \frac{2}{7}t^{7/2} + \frac{6}{5}t^{5/2} + \frac{4}{3}t^{3/2} + C \end{aligned}$$

(b) $\int\left(x^2 + 1 + \frac{1}{x^2+1}\right)dx = \frac{x^3}{3} + x + \tan^{-1}x + C$

(c) $\int\left(\frac{1+r}{r}\right)^2 dr = \int\frac{1+2r+r^2}{r^2}dr = \int\left(r^{-2} + 2r^{-1} + 1\right)dr = -r^{-1} + 2\ln|r| + r + C = -\frac{1}{r} + 2\ln|r| + r + C$

(d) $\int\left(2 + \tan^2\theta\right)d\theta = \int\left[2 + \left(\sec^2\theta - 1\right)\right]d\theta = \int\left(1 + \sec^2\theta\right)d\theta = \theta + \tan\theta + C$

8.

(a) Let $u = x^3 + 1$. Then $du = 3x^2 dx$ and $x^2 dx = \frac{1}{3} du$, so

$$\int x^2 \sqrt{x^3+1} dx = \int \sqrt{u} \left(\frac{1}{3} du \right) = \frac{1}{3} \frac{u^{3/2}}{3/2} + C = \frac{1}{3} \cdot \frac{2}{3} u^{3/2} + C = \frac{2}{9} \left(x^3 + 1 \right)^{3/2} + C.$$

(b) Let $u = \sin\theta$. Then $du = \cos\theta d\theta$, so $\int \sin^2\theta \cos\theta d\theta = \int u^2 du = \frac{1}{3}u^3 + C = \frac{1}{3}\sin^3\theta + C$.

(c) Let $u = 1 + \frac{1}{x}$. Then $du = -\frac{1}{x^2}dx$ and $\frac{1}{x^2}dx = -du$, so $\int \frac{1}{x^2}\sqrt{1+\frac{1}{x}}dx = \int \sqrt{u}(-du) = -\frac{2}{3}u^{3/2} + C = -\frac{2}{3}\left(1+\frac{1}{x}\right)^{3/2} + C$.

9.

(a) Let $u = 1 - x^2$. Then $du = -2xdx$ and $xdx = -\frac{1}{2}du$, so $\int x\sqrt{1-x^2}dx = \int \sqrt{u}\left(-\frac{1}{2}du\right) = -\frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = -\frac{1}{3}\left(1-x^2\right)^{3/2} + C$.

(b) Let $u = 5 - 3x$. Then $du = -3dx$ and $dx = -\frac{1}{3}du$, so

$$\int (5-3x)^{10} dx = \int u^{10} \left(-\frac{1}{3} du \right) = -\frac{1}{3} \cdot \frac{1}{11} u^{11} + C = -\frac{1}{33}(5-3x)^{11} + C.$$

(c) Let $u = -t^4$. Then $du = -4t^3 dt$ and $t^3 dt = -\frac{1}{4}du$, so $\int t^3 e^{-t^4} dt = \int e^u \left(-\frac{1}{4}du\right) = -\frac{1}{4}e^u + C = -\frac{1}{4}e^{-t^4} + C$.

(d) Let $u = 1 + \cos t$. Then $du = -\sin t dt$ and $\sin t dt = -du$, so $\int \sin t\sqrt{1+\cos t}dt = \int \sqrt{u}(-du) = -\frac{2}{3}u^{3/2} + C = -\frac{2}{3}(1+\cos t)^{3/2} + C$.

(e) Let $x = 1 - e^u$. Then $dx = -e^u du$ and $e^u du = -dx$, so

$$\int \frac{e^u}{\left(1-e^u\right)^2} du = \int \frac{1}{x^2}(-dx) = -\int x^{-2} dx = -\left(-x^{-1}\right) + C = \frac{1}{x} + C = \frac{1}{1-e^u} + C.$$

(f) Let $u = 3ax + bx^3$. Then $du = \left(3a + 3bx^2\right)dx = 3\left(a + bx^2\right)dx$, so

$$\int \frac{a+bx^2}{\sqrt{3ax+bx^3}} dx = \int \frac{\frac{1}{3}du}{u^{1/2}} = \frac{1}{3}\int u^{-1/2} du = \frac{1}{3}\cdot 2u^{1/2} + C = \frac{2}{3}\sqrt{3ax+bx^3} + C.$$

(g) Let $u = \ln x$. Then $du = \frac{dx}{x}$, so $\int \frac{(\ln x)^2}{x} dx = \int u^2 du = \frac{1}{3}u^3 + C = \frac{1}{3}(\ln x)^3 + C$.

(h) Let $u = \cos x$. Then $du = -\sin x dx$ and $-du = \sin x dx$, so

$$\int \sin x \sin(\cos x) dx = \int \sin u(-du) = (-\cos u)(-1) + C = \cos(\cos x) + C.$$

Bonus Question

1. Let $g = e^f$.

   (a) Let's first assume that $f'' > 0$. Then $g' = e^f f'$ and $g'' = e^f[f'' + (f')^2]$. e^f is always positive and $(f')^2 \geq 0$. Also $f'' > 0$. Thus $g'' > 0$ and e^f is convex.

   (b) In the most general case, we can argue as follows. Suppose a function h is defined on an interval I. The convexity of f on I is equivalent to the following condition:

   $$h(\lambda x + (1-\lambda)y) < \lambda h(x) + (1-\lambda)h(y) \tag{1}$$

   for any $\lambda \in (0,1)$ and $x, y \in I$. In fact, the left-hand side of (1) is the value of h at $\lambda x + (1-\lambda)y$, and the right-hand side is the value of the secant line connecting $(x, h(x))$ and $(y, h(y))$ at $\lambda x + (1-\lambda)y$. We now use the condition (1) to verify that e^f is convex when f is convex: for any $\lambda \in (0,1)$ and any x, y in the domain of f,

   $$\begin{array}{rll} f(\lambda x + (1-\lambda)y) & < \lambda f(x) + (1-\lambda)f(y) & \text{because } f \text{ is convex.} \\ \text{Therefore } e^{f(\lambda x+(1-\lambda)y))} & \leq e^{\lambda f(x)+(1-\lambda)f(y)} & \text{because } w \mapsto e^w \text{ is increasing,} \\ & < \lambda e^{f(x)} + (1-\lambda)e^{f(y)} & \text{because } w \mapsto e^w \text{ is convex, since its second derivative is positive.} \end{array}$$

   So e^f is convex.

2. Let $f(x) = -\ln x$ and $g(x) = x^2 + 1$. Then $h(x) = (f \circ g)(x) = -\ln(x^2+1)$. $f(x)$ and $g(x)$ are convex since $f''(x) = x^{-2} > 0$ for $x > 0$ and $g''(x) = 2 > 0$ for $x \in \mathbb{R}$. But $h(x) = (f \circ g)(x)$ is not convex as $h''(x) = (2x^2 - 2)(x^2+1)^{-2}$ can be negative when $-1 < x < 1$.